MATH2056 F make up exam-SPR2021 Date:

The GSM phones are not allowed. Notes/boks (only printed!) and the calculators can be used.

All the answers must be clearly stated, otherwise no partial credit will be given.

Please write your answers well-ordered and clearly-visible: this is the responsibility of the student.

The duration is 110 minutes.

When the exam is completed, please scan and upload your answer sheet to UES system:

USE PDF FORMAT.

1. consider the matrix A=$\begin{bmatrix} 1 & \propto & -\gamma & \propto \\ 3\propto & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \propto^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$. Some of the elimination matrices are given below:

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$.

a. (5 pts) Find the unknown entries $\propto, \gamma, \beta, \mathrm{k\ and\ m} =?$
b. (5 pts) Find the eliminated upper triangle matrix U=?
c. (10 pts) Determine the remaining elimination matrice(s).

2. A geometry in 'k' dimensional space is given by x=5y expression, where 'x' and 'y' represents the axis. The intersection with another geometry gives a volume.

a. Give the minimum dimension of the space with your valuable comments (10 pts).
b. Propose an expression for the unknown geometry to have a volume in intersection (10 pts).
c. I proposed a geometry x+4y-z=6. Find all vectors in the intersection. Do these vectors form a vector space? The intersection has how many dimensions inside how many dimensional space? (15 pts)

3. Suppose u is a unit vector in Rn. . This problem is about the n by n symmetric matrix H = I − 2uuT :
(a) (5points) Show directly that H2 = I.
(b) (5 points) Show that H is an orthonormal matrix.
(c) (5 points) One eigenvector of H is u itself. Find the corresponding eigenvalue.
(d) (5 points) Is H diagonalizable? Why or why not?

4. Two tanks are connected, first tank has v(0)=10 liters liquid and the second tank has w(0)=40 liters of water. The liquid diffusion between the tanks are proportional to the difference *v*-*w*.

a)Which one of the following differential equation set expresses the process? (v_k: the liquid diffused to the second tank, w_k: the water diffused to the second tank. 10 pts)

$\begin{bmatrix} v_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix}$ $\quad \frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}$ $\quad \frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}$

b) Determine percentages of the mixture in tank1 at time t=1sec. (10 pts)
c) Find how much liters of liquids are exist in each tank in steady state? (10 pts)